MANCHESTER 1824
The University of Manchester

AERO60492 – Autonomous Mobile Robots

Lecture 14 – Indoor Sensing for Aerial Robots

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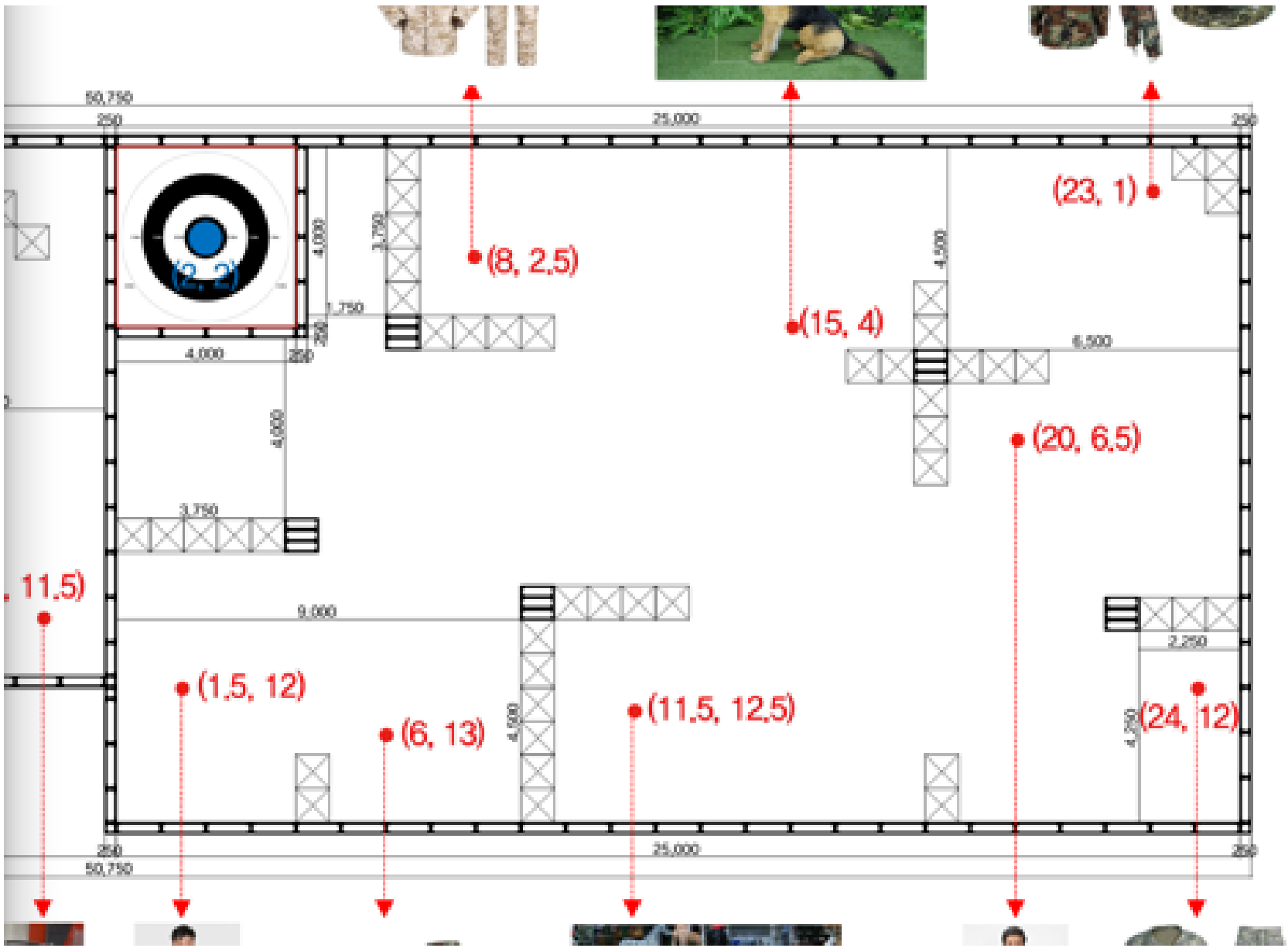
Intended Learning Outcomes

- To understand limitations of existing localization and mapping sensors indoors
- To appreciate existence of indoor positioning systems
- To appreciate SLAM and its importance to mobile robot
- To understand general principle behind SLAM
- To summarize and be able to utilize information given to you about different locomotion, sensors and algorithms for robot design

Sample missions in indoor environment



The need for Localisation

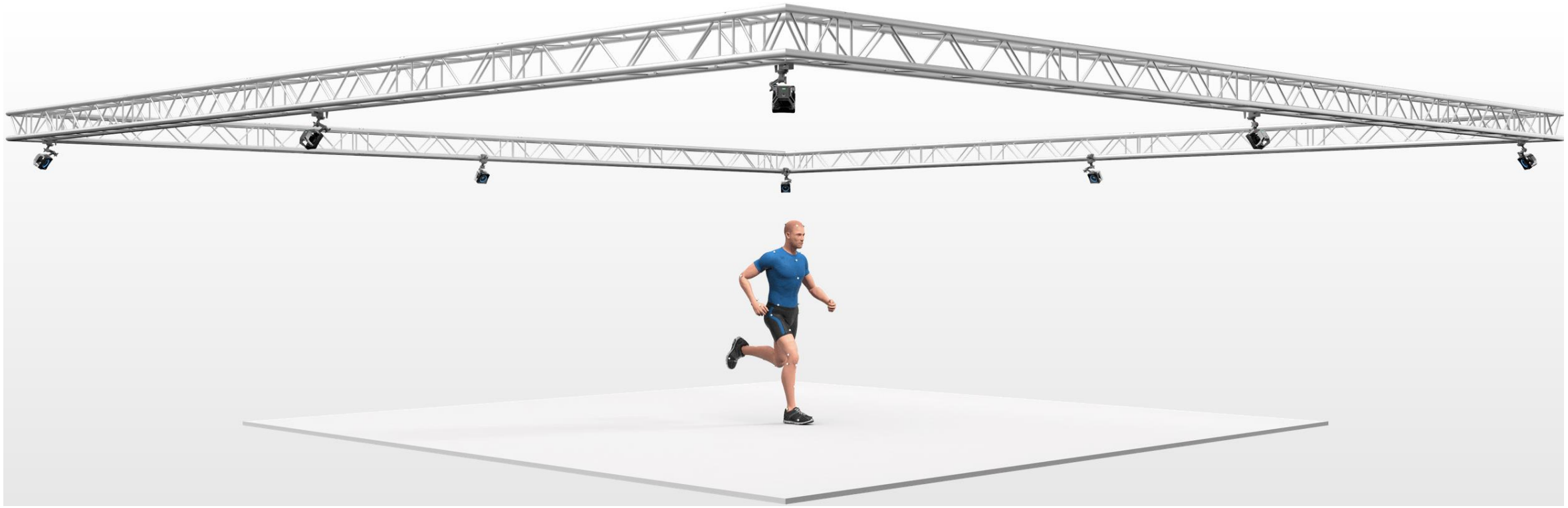
- To localise rooms, and be able to navigate itself to specific points
- To localise objects and mission critical items
- To understand where given measurement was taken at
- To avoid obstacles and keep track where the robot has been

Localisation using GPS

- GPS cannot be used indoors:
  - Signal blocked by walls
  - Error too large
- The need for indoor navigation methods arise
  - Motion capture
  - Markers
  - Range finder systems

Localisation using motion capture/external systems

- Used in entertainment industry:



Localisation using motion capture/external systems

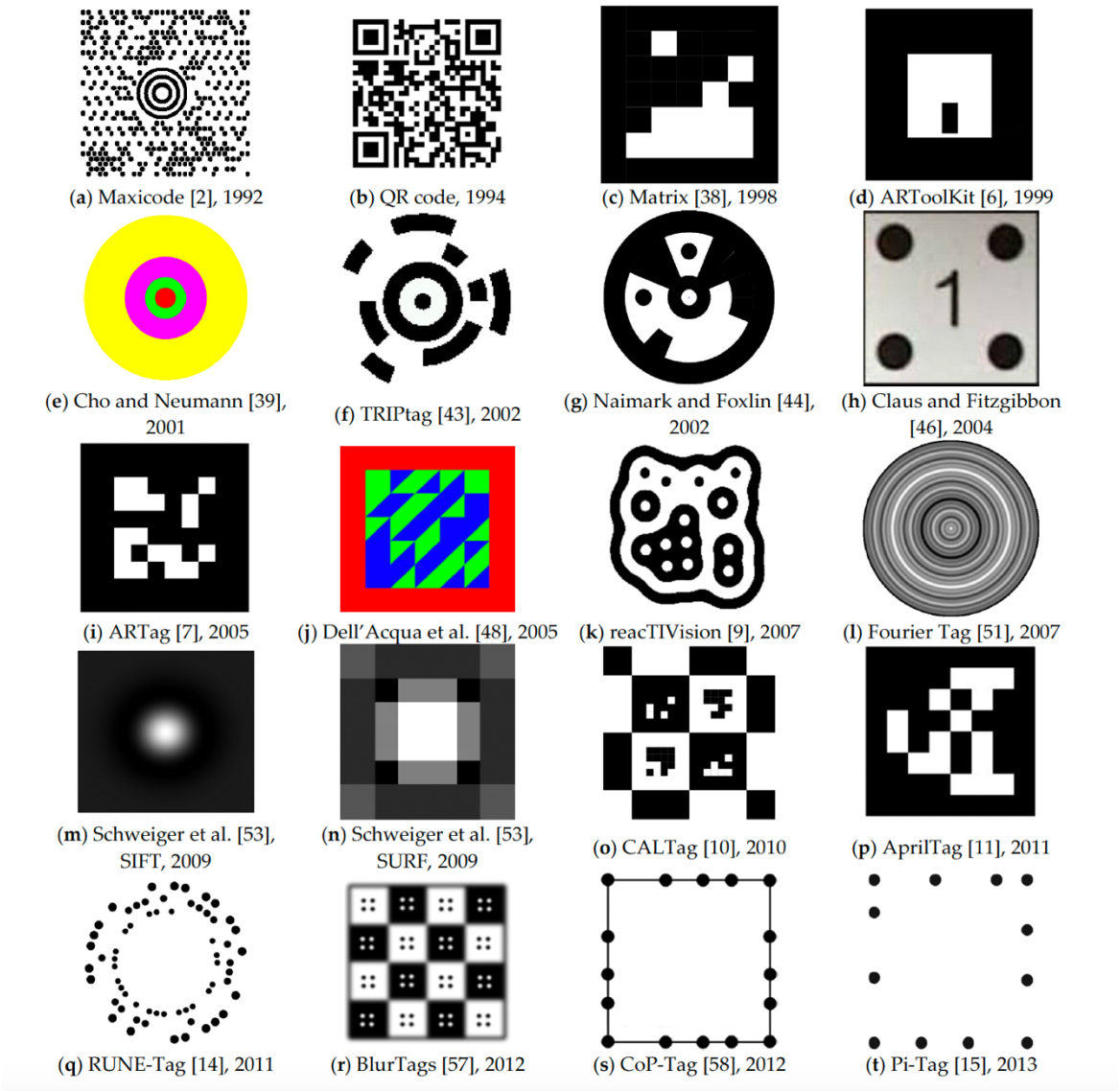
Pros	Cons
Extremely accurate < 1 mm and very fast more than 100 Hz positional update	… but unrealistic when testing other methods
Easy to add more vehicles	Requires Line of Sight
	Expensive
	Markers must not move
	Requires camera setup

Other base station methods

- SteamVR Base Station 2.0
- UWB systems
- And others…

Fiducial Markers

- They are designed to allow easy computation of relative position in the image frame.
- They can be placed on the walls and ground, each identifying specific position
- Robot can read them using camera and determine its position relative to known tag



Localisation using Fiducial markers pros and cons

Pros	Cons
Accurate	Requires Line of Sight
Cheap	Markers must not move
Can be used for arbitrary number of vehicles	Requires marker setup and careful measurement

- The need for fully robot-internal (no external setup) localisation system arises

The need for mapping

- Most buildings have maps based on plans, however they can be out of date:
  - The building was modified in some way
  - It was simply not built up to spec
  - It got damaged during natural disaster
- Also, most maps do not account for furniture etc.
- Thus the robot needs to map its environment too
- Aforementioned methods do not provide any mapping

Combining both

- You cannot use onboard sensors localization without accurate map
- You cannot create accurate map without accurate localization.
- We can do them together!
- Hence Simultaneous Localisation and Mapping (SLAM) algorithm was born

Formal definition of SLAM

- The SLAM objective is to find:

$$p(\boldsymbol{x}_k, \boldsymbol{m} | \boldsymbol{Z}_{0:k}, \boldsymbol{U}_{0:k}, x_{0:k-1})$$

Current position

The set of all landmarks

The set of all observations

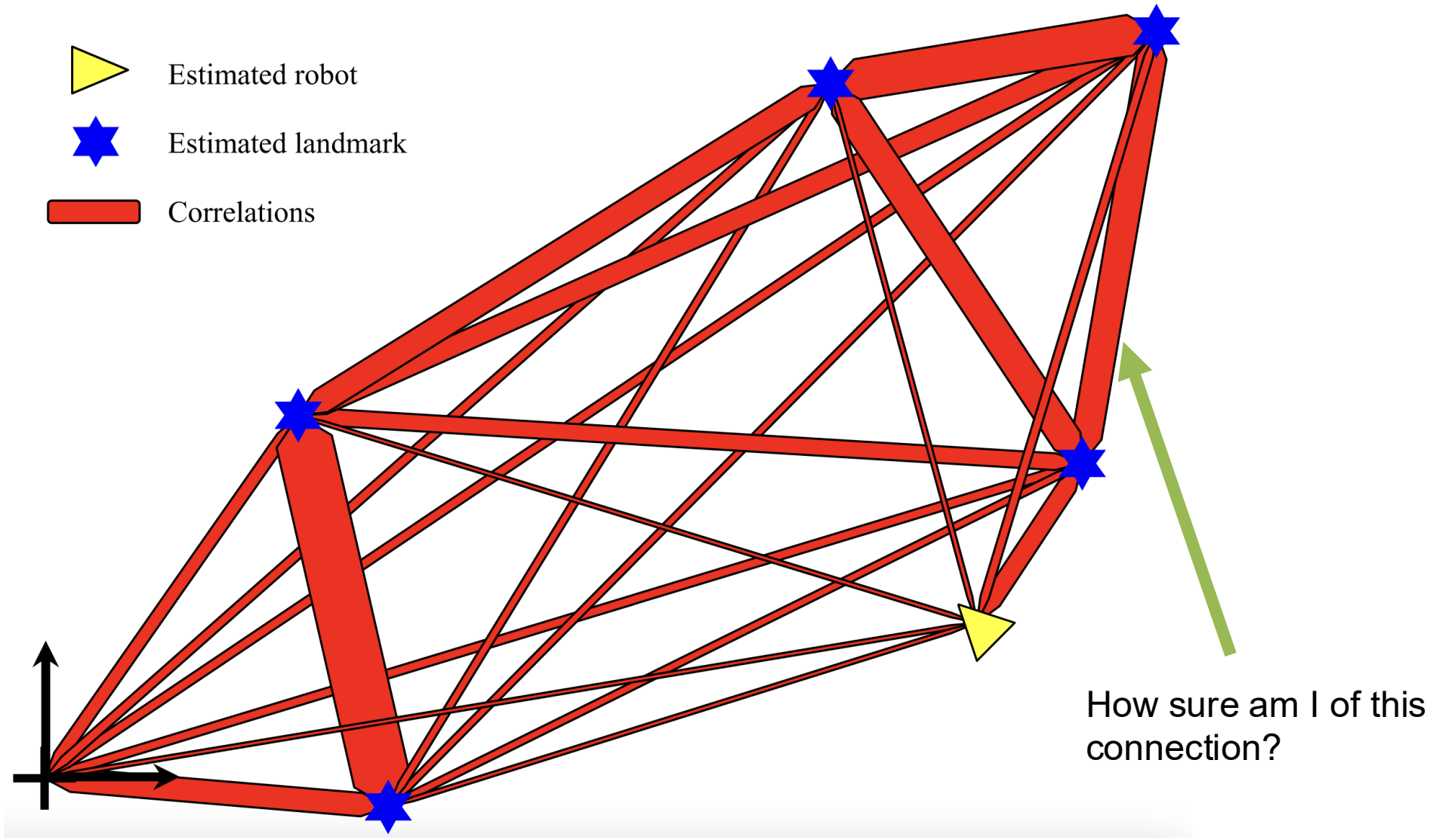
The set of all commands

The initial/past positions

What are landmarks?

- Imagine you are trying to explore new environment.
- Naturally, you would find some characteristic features (such as unusual building, shape, color…) to help you navigate.
- Robots fundamentally try to replicate that
- Landmark in SLAM could be
  - Corner
  - Door gap
  - Table leg etc.

SLAM pictographically represented



SLAM features

- Can work with arbitrary sensors, although LIDAR is most commonly used
- Can be 2D or 3D depending on sensor type – 3D computationally expensive
- Necessary for truly autonomous robots
- Struggles with moving objects

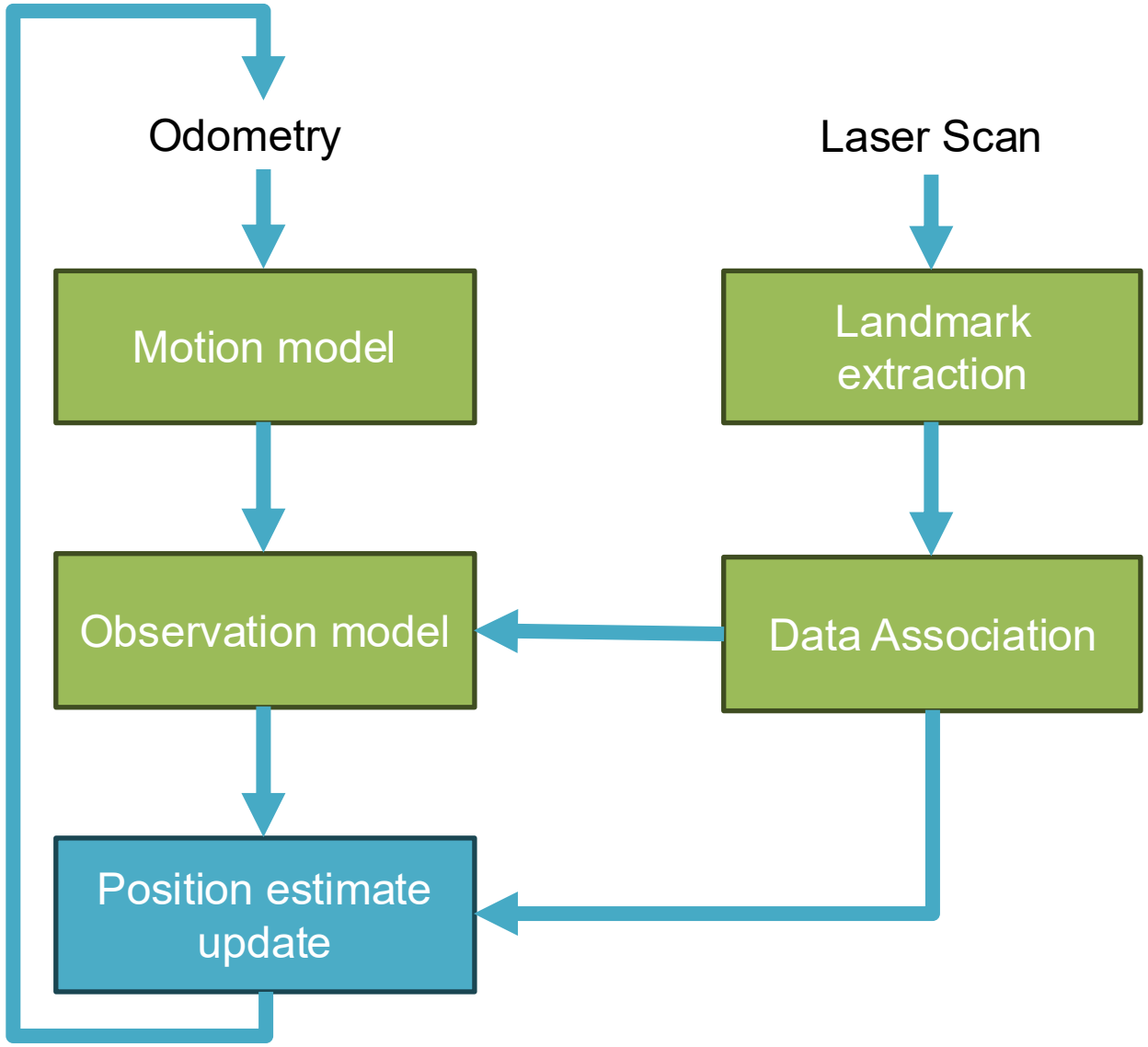
SLAM sensors requirements

- LIDAR 2D/3D – 3D is more commonly used in Aerial robots
- IMU – To give idea how far robot has moved
- Camera 2D/3D camera – you can use it for Visual-SLAM
- Wheel encoders – if present
- We can often combined LIDAR-SLAM and Visual-SLAM to get better performance

SLAM applications

- SLAM is applicable to all robotic types: air, ground, water
- Example applications:
  - Lawn mowers
  - vacuum cleaners
  - Aerial inspections
  - Underground exploration
  - Underwater exploration
  - Localisation and mapping of distant planets

SLAM Overview



Motion model

- Describes how much robot has moved since last time step
- It is not accounting for landmarks and it is just estimation
- Formally it is defined as:

$$p(x_k | x_{k-1}, u_{k-1})$$

Current position

Previous position

The control command

Observation model

- Describes likelihood of observations given robot position and landmarks
- Formally defined:

$$p(z_k | x_k, \boldsymbol{m})$$

Observation Current position Landmark list

- If robot thinks it moved where it moved, how likely it is to see current observation?

Landmark extraction

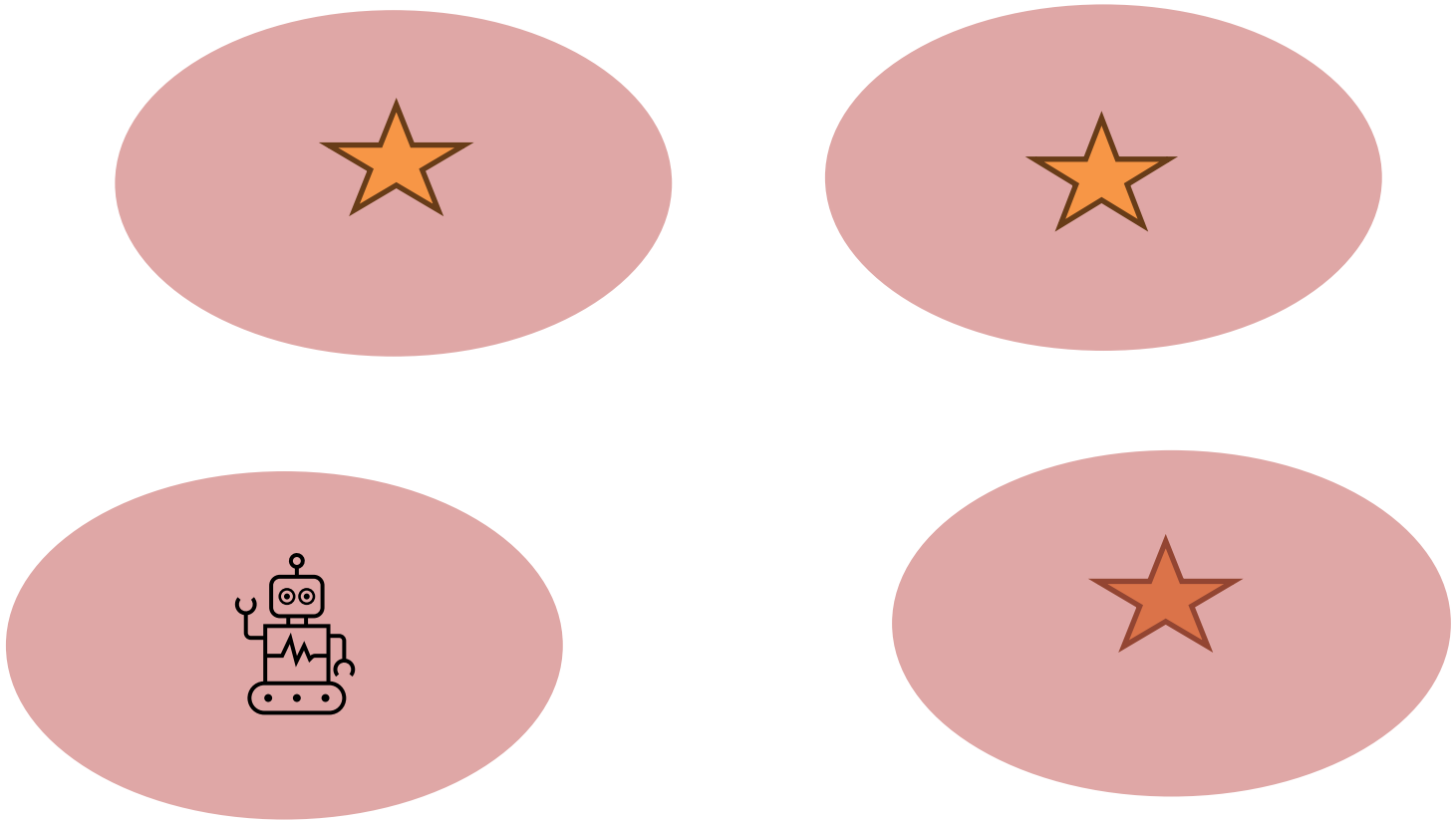
- Good landmark has following properties:
  - Re-observable
  - Distinguishable from the environment
  - Stationary
  - Plentiful
- Some of the most popular landmark extraction:
  - Spike, just look for spike in your data
  - RANSAC (Random Sampling Consensus) – extract straight lines (such as walls)

Data association

- Answers question: Have we seen this landmark before or is this brand new landmark?
- In essence three problems can arise:
  - not see any landmark at given time step
  - observe landmark and never see it again
  - Wrongly associated landmark
- Data association algorithm needs to cope with those problems for reliable SLAM!

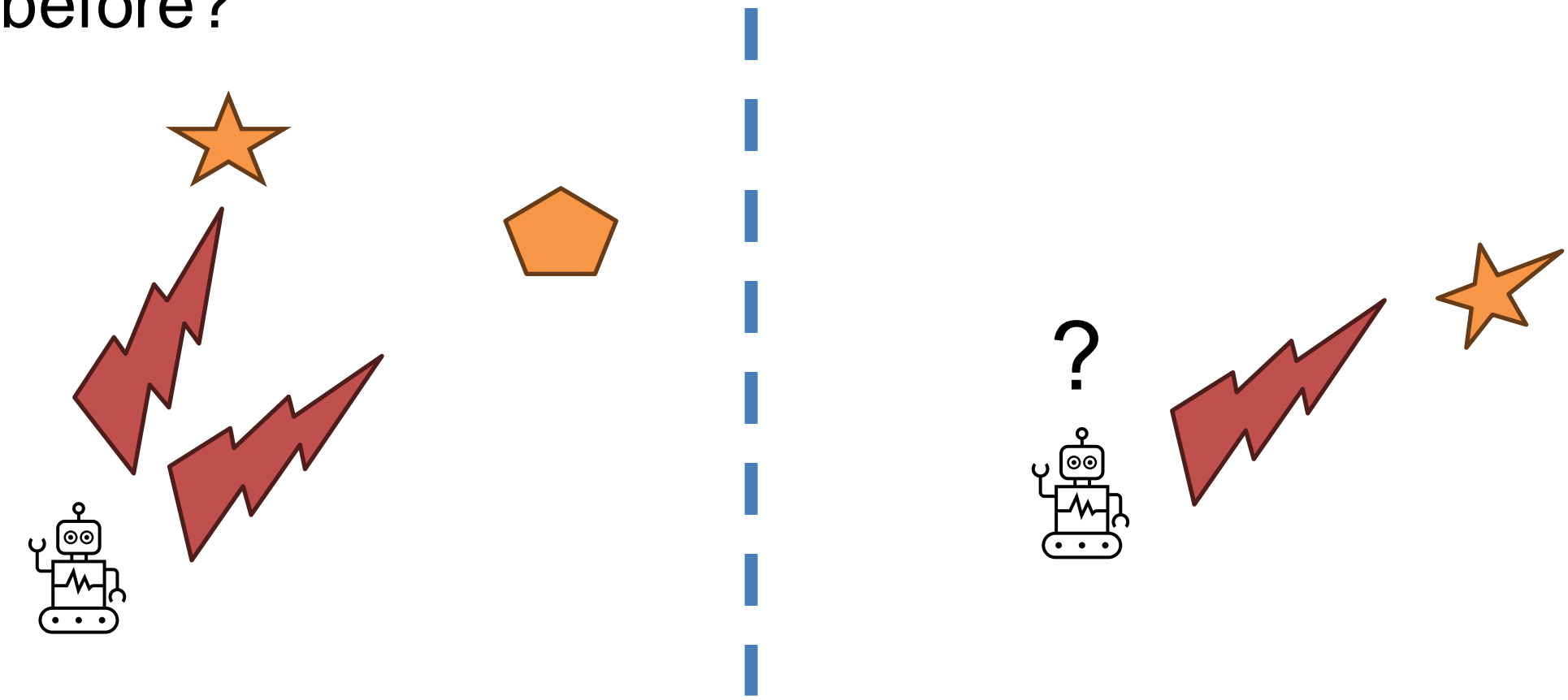
SLAM Challenges: 1

- Positions of both robot and landmark are inherently uncertain:



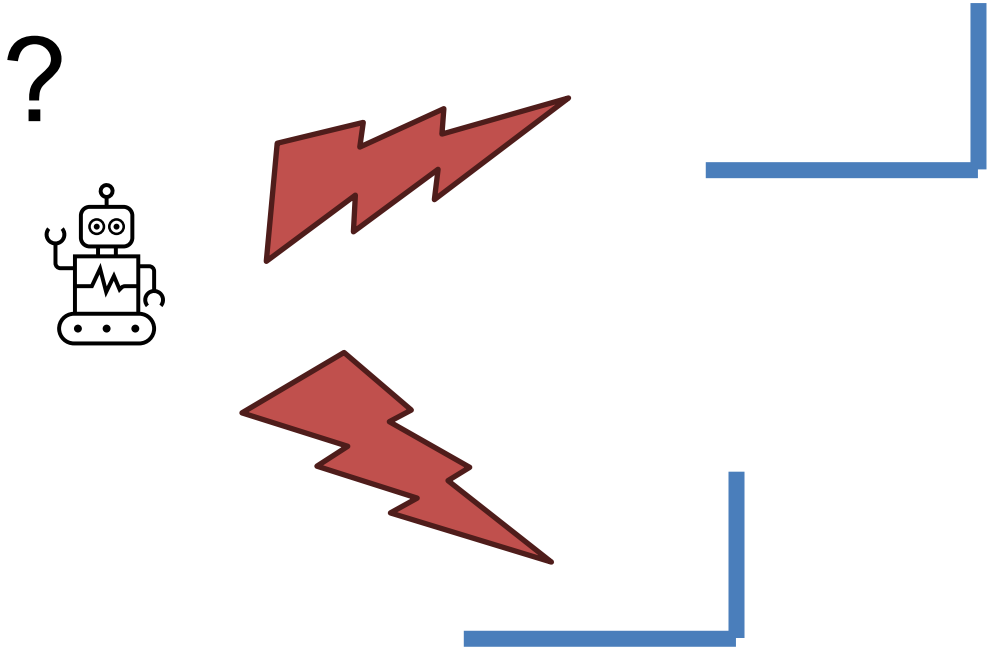
SLAM Challenges 2

- Landmark association, Have this landmark been seen before?



SLAM Challenges 3

- Misleading landmarks – landmarks which look the same but at different positions



Summary

- Indoor localization methods were discussed
- Brief introduction to SLAM was given
- SLAM definition was formalized
- Limitations of SLAM were discussed

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Quiz